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SEEING CAUSINGS AND HEARING GESTURES

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Can humans see causal interactions? Evidence on the visual perception of causal interactions, from Michotte to contemporary work, is best interpreted as showing that we can see some causal interactions in the same sense as that in which we can hear speech. Causal perception, like speech perception, is a form of categorical perception.

Which properties and happenings can humans see? It is fairly uncontroversial that we can see the shapes of things and their movements. It is also reasonably uncontroversial that we cannot see properties like market value or processes like radioactive decay. People may sometimes talk about seeing these things, but it seems plausible that what they strictly and literally see are only characteristic visual indicators of market value or radioactive decay.

What about simple causal interactions like launchings, burstings, blockings and supportings – can we see these things? When two balls collide, what do we see? Do we see only one motion followed by another, or do we see some kind of causal interaction?

David Hume appears to have held that we perceive only one motion followed by another: 'When we consider these objects with the utmost attention, we find only that one body approaches the other; and that the motion of it precedes that of the other, but without any sensible interval' (*Treatise* I iii 2). Albert Michotte, on the other hand, held that we can see certain types of causal interaction in the same sense as that in which we can see shape or movement:

> There are some cases ... in which a causal impression arises, clear, genuine, and unmistakable, and the idea of cause can be derived from it by simple abstraction in just the same way as the idea of shape or movement can be derived from the perception of shape or movement.[1]

Several philosophers and psychologists agree with Michotte that causal interactions are visible in the same sense as shape and motion are. Among

[1] A. Michotte, *The Perception of Causality*, tr. T. Miles (London: Methuen, 1946), pp. 270–1.

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the philosophers are David Armstrong[2] and Peter Strawson, who says

> In a great boulder rolling down the mountainside and flattening the wooden hut in its path we see an exemplary instance of force ... these mechanical transactions ... are directly observable (or experienceable).[3]

The psychologists Brian Scholl and Patrice Tremoulet agree that we can see causal interactions:

> ... just as the visual system works to recover the physical structure of the world by inferring properties such as 3-D shape, so too does it work to recover the causal ... structure of the world by inferring properties such as causality.[4]

While agreeing that we can see some causal interactions, Michotte, Strawson and Scholl adopt different approaches to thinking about perception. Scholl is driven by David Marr's question 'What kind of information is vision really delivering?'.[5] Strawson focuses on descriptions of perceptual content which describe how the world is presented to the perceiver. Michotte is concerned with perception as a source of causal concepts.

The variety of approaches to perception can easily complicate the task of answering questions about what we can perceive. What could ensure that in making claims about what can or cannot be seen, we are genuinely agreeing with or contradicting others who make similar sounding claims, rather than merely using terms like 'see' in a way which fails to connect with other positions? Michotte, Strawson and Scholl all hold that we can see causal interactions in the same sense as we can see shape, whatever sense that is. The appeal to shape perception ensures genuine agreement, despite any differences about the nature of perception.

In this paper I argue that we can see some causal interactions in the same sense as we can hear speech, whatever sense that is. I shall also tentatively suggest that speech perception is importantly different from shape perception. To see the shapes of things is, in a wide range of naturally occurring conditions if not always, to be aware of those shapes and to be in a position to know things about them; but hearing speech and seeing causings are not similarly related to awareness of the things perceived. If so, Michotte, Strawson and Scholl are right that we can see some causal interactions, but wrong that we can do so in the same sense as we can see shape.

[2] D.M. Armstrong, 'Going through the Open Door Again', in J. Collins *et al.* (eds), *Causation and Counterfactuals* (MIT Press, 2004), pp. 445–58.

[3] P.F. Strawson, 'Causation and Explanation', in his *Analysis and Metaphysics* (Oxford UP, 1992), p. 118.

[4] B.J. Scholl and P.D. Tremoulet, 'Perceptual Causality and Animacy', *Trends in Cognitive Sciences*, 4 (2000), pp. 299–309, at p. 299.

[5] D. Marr, *Vision: a Computational Investigation into the Human Representation and Processing of Visual Information* (San Francisco: W.H. Freeman, 1982), p. 35.

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I shall only consider visually presented stimuli. Some researchers hold that we perceive causal interactions through touch or experiences of agency. These views will not be considered here. Following common use, I adopt 'causal perception' as a label for whatever perceptual experiences are associated with visual stimuli involving or standing in for causal interactions. Understood in this way, my first question is whether the objects of causal perception are causal interactions or just movements.

I

Although we have quite a lot of data on causal perception, the interpretation of these data is not straightforward.

I shall take Albert Michotte's experiments on launching as my starting-point. The launching sequence is illustrated schematically in Figure 1.[6] When seeing this sequence, most subjects describe their experiences in causal terms as the experience of a collision or launching. They also distinguish this experience from the experience of seeing one movement followed by another. (The latter experience is reported when there is a delay between the two movements.) Michotte studied launching in great detail, and I shall discuss his findings below.

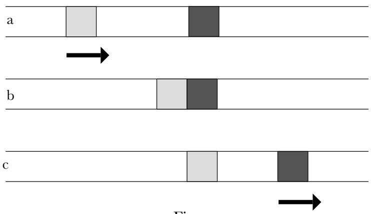

Figure 1

Peter White and Elizabeth Milne showed subjects various other kinds of schematic animation (see Figure 2). They found that subjects reliably reported seeing pulling, disintegration and bursting.[7] There is no doubt, then, that people have distinctive experiences for a range

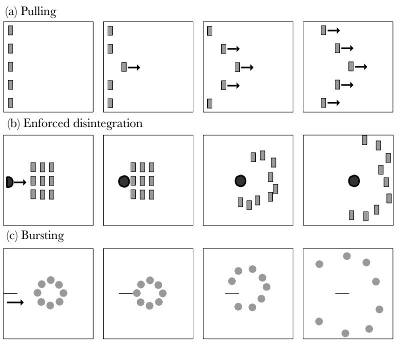

Figure 2

[6] A video is available on Brian Scholl's website, http://pantheon.yale.edu/~bs265/demos/causality.html. The figure is from G. Thines, A. Costall and G. Butterworth (eds), *Michotte's Experimental Psychology of Perception* (Hillsdale: Erlbaum, 1991), p. 69.

[7] P. White and E. Milne, 'Impressions of Enforced Disintegration and Bursting in the Visual Perception of Collision Events', *Journal of Experimental Psychology: General*, 128 (1999), pp. 499–516, and 'Phenomenal Causality: Impressions of Pulling in the Visual Perception of Objects in Motion', *American Journal of Psychology*, 110 (1997), pp. 573–602. The illustration is from Scholl and Tremoulet, 'Perceptual Causality and Animacy', p. 301.

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of stimuli which mimic different types of causal interaction, and that people regularly describe these experiences in causal terms.

Elizabeth Anscombe, Curt Ducasse and Peter Strawson have all observed that there are also many everyday situations in which it is natural to describe what we see in causal terms.[8] These include seeing a bird bending a branch by alighting on it and a boulder crushing a hut as it rolls down a mountainside. Reading these authors, it may seem tempting to conclude from what people unreflectively say about their experiences that we can perceive causal interactions.

I do not think this conclusion follows. When shown an animation involving simple geometric shapes of the sort used by Heider and Simmel, many people report seeing things such as the large triangle intimidating the small one, or the small triangle defending the little circle as the big triangle tries to attack it.[9] (Figure 3 shows one frame from Heider and Simmel's animation.) This suggests that there may be nothing that people will not speak of seeing, given the right conditions. So if there is a substantial question about whether we can see some types of causal interaction, it cannot be correct to answer this question on the basis of what people say they see.

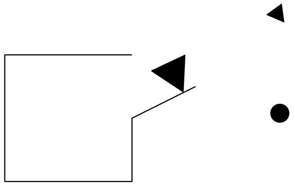

Figure 3

I am not suggesting that people who talk about seeing intimidating or defending are mistaken or confused, only that common sense talk about perception serves a range of practical interests which probably do not include, and are in any case not limited to, answering theoretical questions about the nature of vision. We should not take what people say about their perceptions at face value, any more than what they say about the meanings of their utterances.

In cases like intimidating or defending it seems plausible to say that we see, at most, visible indicators of these things, not the interactions themselves. While there may be special phenomenological effects characteristic of intimidating or defending, it does not follow that we can see these things happen (and of course Heider and Simmel do not claim that we can). This is

[8] G.E.M. Anscombe, 'Causality and Determination', in E. Sosa and M. Tooley (eds), *Causation* (Oxford UP, 1993), pp. 88–104; C.J. Ducasse, 'Causation: Perceivable? Or Only Inferred?', *Philosophy and Phenomenological Research*, 26 (1965), pp. 173–9; Strawson, *Analysis and Metaphysics*, pp. 109–32.

[9] F. Heider and M. Simmel, 'An Experimental Study of Apparent Behaviour', *American Journal of Psychology*, 57 (1944), pp. 243–9, figure at p. 244. At the time of writing there is a reconstruction of Heider and Simmel's animation online at http://anthropomorphism.org/img/Heider_Flash.swf.

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not to say that future research could not show that subjects really do see intimidating, only that nothing so far considered supports this conclusion. Those who follow Hume in denying that we can perceive causal interactions can say the same thing about findings like Michotte's and White and Milne's. They can admit that various stimuli give rise to characteristic phenomenology which many people unreflectively describe in causal terms, yet deny that we experience causal interactions.

Some philosophers seem to have missed this point. In ch. 4 of his *Intentionality*, John Searle objects to an analysis of causation in terms of regularities on the ground that it 'flies in the face of our common sense conviction that we do perceive causal relations all the time'. Searle may be wrong to suppose that there is any such common sense conviction; for reasons just given he is certainly wrong to suppose that such a conviction would support a claim about what we perceive. Searle then notes that 'The experience of perceiving one event following another is really quite different from the experience of perceiving the second event as caused by the first'.[10] By itself, this observation is unhelpful, because something similar is true of experiences of perceiving intimidating or defending. Searle's final point is that 'the researches of Michotte and Piaget would seem to support our common sense view'. As I shall show by taking a closer look at Michotte, this is not exactly straightforward. Michotte himself acknowledges this. At one point (*The Perception of Causality*, p. 256) he imagines Hume learning about his experiments, and writes 'it is probable that his philosophical position would not have been affected in the least'. In short, neither Michotte's nor more recent research (more of which is considered below) straightforwardly shows that we can see causings. In order to defend the claim that we can see causal interactions, one needs a way of getting beyond what some people unreflectively say about their experiences.

II

How can one get past disagreement over unreflective descriptions of visual experiences? By finding something that needs explaining.

Susanna Siegel suggests considering carefully matched pairs of experiences. For example, she imagines two ways of experiencing a situation in which a ball lands in a plant pot just before the lights go out.[11] You might find it compelling to suppose that the ball's landing causes the lights to go

[10] J.R. Searle, *Intentionality: an Essay in the Philosophy of Mind* (Cambridge UP, 1983), pp. 114–15.
[11] S. Siegel, 'The Visual Experience of Causation', *The Philosophical Quarterly*, 59 (2009) (this issue), pp. 519–40.

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out (even though you would presumably not judge this to be so), or you might experience the two events as unconnected. Siegel claims that these are two distinct experiences to which the same stimuli can give rise. Her question is then what explains the difference between the two experiences. She argues that the difference is best explained by supposing that the two events are experienced as causally connected in one case, but not in the other.

Siegel's 'matched pairs' approach looks promising because it identifies something that needs explaining, a difference between two experiences. But can Siegel's approach be applied to Michotte's launching stimulus? There would have to be a pair of occasions on which a subject saw the same type of launching sequence, experiencing successive movements on one occasion and experiencing launching on the other. Unfortunately, it seems there are no such pairs, because in the case of launching and similar stimuli, the characteristic phenomenology described as causal appears to be mandatory if you are paying attention. So one cannot apply Siegel's approach here.

This problem relates to a defect of Siegel's approach: it does not distinguish perceptual phenomenology from the non-perceptual phenomenology associated with the feeling of being struck by an idea. When a ball drops just as a light goes out, are we struck by the thought that the one caused the other, or do we somehow perceive this? Siegel's approach does not distinguish these two possibilities, which undermines her claim to have shown that 'causation is represented in visual experience'. When Michotte (p. 257) discusses the sort of cases Siegel focuses on, he denies that causation is perceived, but notes that 'a causal interpretation is urgently called for'. This seems at least as plausible as Siegel's claim that we perceive a causal interaction when a ball lands in a plant pot just before the lights go out. Indeed, it is tempting to think that genuinely perceptual phenomenology is always mandatory for subjects attending to the relevant aspects of stimuli. If so, the only phenomenological effects Siegel's 'matched pairs' approach can discern are non-perceptual, because the required pairs of experiences do not exist when the phenomenology is mandatory.

My aim is to support the claim that some causal interactions can be seen by finding something that needs explaining. If Siegel's matched pairs of experiences do not provide a suitable target for explanation, what else might?

Brian Scholl and Ken Nakayama offer a promising candidate which needs explaining: illusory crescents.[12] Shown a display just like Michotte's launching stimulus, except that the two objects completely overlap before the second starts to move,[13] almost all subjects report seeing a single object

[12] B.J. Scholl and K. Nakayama, 'Causal Capture: Contextual Effects on the Perception of Collision Events', *Psychological Science*, 13 (2002), pp. 493–8.

[13] A video is available on Scholl's web site referred to in fn. 6 above.

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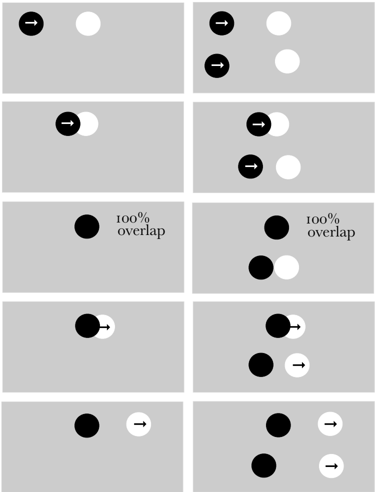

Figure 4

moving in front of or behind a stationary object; some report that the moving object changes colour (for a standard overlap event, see the left-hand side of Figure 4). However, it is possible to enhance this display so as to induce subjects to report seeing the overlap event as a launching event rather than as the movement of a single object (as in the overlap event with contextual launching event on the right-hand side of Figure 4). Here is the fact to be explained: when subjects report seeing the overlap event as launching, they also report seeing an illusory crescent, as if the shapes did not completely overlap; and they systematically underestimate the degree of overlap between the two objects.[14]

Does explaining this fact require conceding that we can see some types of causal interaction? I shall allow that as Scholl and Nakayama claim, subjects are seeing illusory crescents, and that the experience characteristic of launching affects the perception of shape.[15] Given this, Scholl and Nakayama offer (p. 466) 'a simple categorical explanation for the causal crescents illusion':

> ... the visual system, when led by other means to perceive an event as a causal collision, effectively 'refuses' to see the two objects as fully overlapped, because of an internalized constraint to the effect that such a spatial arrangement is not physically possible. As a result, a thin crescent of one object remains uncovered by the other one – as would in fact be the case in a straight-on billiard-ball collision.

If this is the best explanation of the illusory crescents, then we can indeed see at least one type of causal interaction.

This argument appears to be a promising way of attacking the idea that the experience characteristic of launching involves only some special phenomenological effect indicative of collisions. After all, why would such a phenomenological effect be associated with the illusory perception of crescents? Scholl and Nakayama's position is well supported in so far as

[14] Scholl and Nakayama, 'Illusory Causal Crescents: Misperceived Spatial Relations Due to Perceived Causality', *Perception*, 33 (2004), pp. 455–69, at p. 458.

[15] Cf. Scholl and Nakayama, p. 467: 'the perception of causality can also affect other types of visual processing – in this case the perception of spatial relations among moving objects'.

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explaining the association between seeing launchings and seeing crescents requires appeal to some state with causal content.

But must that state be perceptual? On the face of it, Scholl and Nakayama's finding is equally well explained by either of two hypotheses: (a) there is a visual perception of a collision; or (b) there is some non-doxastic but also non-perceptual cognition of a collision. Since there may well be ways to exclude (b), I do not claim that Scholl and Nakayama's argument is incorrect. But I cannot make use of this argument here without ruling out (b), and that seems to require understanding how perception is distinct from and relates to other forms of cognition. Also, Scholl and Nakayama's strategy may not generalize to cases other than the launching stimuli. It would be better to be able to say more generally which types of causal interactions can be perceived, rather than just that at least one type is perceived (for examples, see White and Milne's stimuli above). For these reasons I shall introduce and then pursue a third and final approach to showing that we can see some types of causal interaction. (The illusory crescents finding will be useful later.)

The Launching and Triggering Effects
Breakdown of stages according to time-interval
(30 readings at each interval)
Speed of A = 40 cm/s; speed of B = 11 cm/s. Ratio 3.6:1

Intervals (ms)	Direct launching
14	100%
28	100%
42	100%
56	100%
70	83%
84	58%
98	50%
112	17%
126	3%
140	0
154	0
168	0
196	0
224	0

Michotte's work provides the most promising candidate for a fact in need of explanation. The first phase of his work involves discovering the precise conditions under which the launching effect occurs, such as the relative speeds of the two objects, the delay between the first and second objects' movements, and the trajectories of the two objects. For instance, what happens if there is a pause between the first object's movement and the second object's movement? How long can the pause be before the launching effect vanishes? Michotte's findings are presented in the adjoining table.[16]

Contemporary philosophers and psychologists sometimes write as if all Michotte did was study, in minute detail, the conditions under which the launching effect occurs. But this is to ignore the crucial second phase of Michotte's work. He was not directly concerned to know that pauses longer than 70 ms dramatically reduce the impression of launching. His question was *why* the launching impression occurs under precisely these conditions and not others. He writes that after identifying conditions under which the

[16] Michotte, *The Perception of Causality*, p. 115, part of table IX.

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launching effect occurs, '... we were confronted with a second task, that of "understanding" the phenomenon, of ... seeking to find out why such and such conditions were necessary for its production, and why it possessed such and such properties' (p. 18).

Michotte's idea, then, is that we can get past superficial ideas about the perceptual experiences characteristic of launching by first (a) identifying the precise conditions under which these experiences occur, and then (b) explaining why they occur under these conditions. If it turns out that the best explanation at stage (b) requires the claim that we see causal interactions, that will constitute an argument for this claim.

The key fact to be explained, then, is that the experience characteristic of launching occurs under certain conditions. Having come this far with Michotte, I propose now to deviate from his views. To develop the argument that we can see causal interactions, I shall compare causal perception with speech perception.

III

One of the questions set for this symposium is what methods we can use to determine the admissible contents of experience. One way to answer this question is to study cases where researchers have been successful, and speech is one of those cases.

According to Liberman and Mattingly's 'motor theory' of speech perception, when we encounter speech we perceive not sounds but gestures. More exactly, we perceive 'the intended phonic gestures of the speaker'.[17]

In the case of causation, the question is whether we perceive merely a sequence of movements, or causal interactions. In the case of speech, the parallel question is whether we perceive sounds or intended phonic gestures.

Naturally, not everyone accepts the motor theory of speech perception.[18] For my purposes, it is most important that Liberman and Mattingly have a

[17] What does 'intended' mean here? It is used to indicate that what are perceived are motor commands which produce gestures rather than gestures themselves. See A.M. Liberman and I.G. Mattingly, 'The Motor Theory of Speech Perception Revised', *Cognition*, 21 (1985), pp. 1–36, at p. 23: 'gestures do have characteristic invariant properties ... though these must be seen, not as peripheral movements, but as the more remote structures that control the movements. These structures correspond to the speaker's intentions.' So far these 'remote structures' are hypothetical: see A.M. Liberman and D.H. Whalen, 'On the Relation of Speech to Language', *Trends in Cognitive Sciences*, 4 (2000), pp. 187–96, at p. 195.

[18] A review of objections and some recent supporting evidence can be found in S. Pinker and R. Jackendoff, 'The Faculty of Language: What's Special About It?', *Cognition*, 95 (2005), pp. 201–36, §2.2. Liberman and Mattingly themselves emphasize that arguments for the theory are not conclusive and further research is needed: Liberman and Mattingly, 'A Specialization for Speech Perception', *Science*, 243 (1989), pp. 489–94, at p. 493.

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valid argument for the conclusion that the objects of speech perception are intended gestures; whether the premises of this argument are true is less important. After all, even if their claims about speech were false, parallel claims might be true of causal perception.

Liberman and Mattingly's argument starts from the premise that speech perception is categorical. To illustrate, Figure 5 represents thirteen sounds spreading across *ba*, *da*, *ga*.[19] Each sound differs from its neighbours by the same amount as any other sound, at least when difference is measured by frequency. Most people would not be able to discriminate two adjacent sounds, except for two special cases (one around –3 to –1 and one around +1 to +3) where discriminating is easier: here people hear the sound change from *da* to *ga* or from *ga* to *ba*. This pattern of heightened discrimination is the defining characteristic of categorical perception.[20] Small changes to stimuli can make large differences to perception, large changes to stimuli can make small differences to perception, and the stimuli can be ordered and sorted into categories; discriminating nearby pairs of stimuli on either side of a category boundary is dramatically easier than discriminating pairs from within a category.

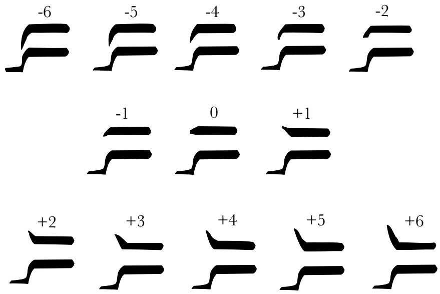

Figure 5 ba-da-ga

The existence of these category boundaries is specific to speech perception as opposed to auditory perception generally. When special tricks are used to make subjects perceive a stimulus first as speech and then as non-speech, the locations of boundaries differ between the two types of perception.[21] Speech perception also exhibits constancy (otherwise called invariance): the location of the category boundaries changes depending on contextual factors such as the speaker's dialect,[22] or the rate at which the speaker talks;[23] both factors dramatically affect which sounds are produced.

[19] Ba-da-ga, from http://www.columbia.edu/itc/psychology/rmk/T2/T2.2b.html.

[20] S. Harnad, 'Psychophysical and Cognitive Aspects of Categorical Perception: a Critical Overview', in S. Harnad (ed.), *Categorical Perception: the Groundwork of Cognition* (Cambridge UP, 1987), pp. 1–28.

[21] See Liberman and Mattingly, 'The Motor Theory of Speech Perception Revised', pp. 20–1.

[22] B.H. Repp and A.M. Liberman, 'Phonetic Category Boundaries Are Flexible', in Harnad (ed.), *Categorical Perception*, pp. 89–112.

[23] L.C. Nygaard and D.B. Pisoni, 'Speech Perception: New Directions in Research and Theory', in J. Miller *et al.* (eds), *Speech, Language and Communication* (London: Academic Press, 1995), pp. 72–5.

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This means that in two different contexts, different stimuli may result in the same perceptions, and the same stimulus may result in different perceptions.[24]

Later I shall defend a distinction between categorical and other forms of perception. It is essential for drawing this distinction that I have characterized categorical perception in terms of abilities to distinguish stimuli, and not in terms of how stimuli appear to perceivers. For example, I have not talked about pairs of sounds being perceived as similar. In my view, which I take to be the standard view, categorical perception is not a matter of how things appear at all.[25]

Given that speech perception is categorical and exhibits constancy, we can ask where the category boundaries fall. Which features of the stimuli best predict category membership? Liberman and Mattingly show that in the case of speech, category boundaries typically correspond to differences between intended phonic gestures. The existence of category boundaries and their correspondence to intended phonic gestures needs explaining.

Following Liberman and Mattingly, one can explain this by postulating a module for speech perception. Anything which is potentially speech (including both auditory and visual stimuli) is passed to the module, which attempts to interpret it as speech. It does this by attempting to replicate stimuli by issuing the same gestures as are also used for producing speech (this is the 'motor' in 'motor theory'). Where a replication is possible, the stimuli are perceived as speech, further auditory or visual processing is partially suppressed, and the module identifies the stimuli as composed of the gestures which were used in the successful replication. Accordingly one can say that the stimuli are perceived as a sequence of phonic gestures.

One line of response to this argument involves attempting to show that the category boundaries correspond to some acoustic property of speech at

[24] Liberman and Mattingly, 'The Motor Theory of Speech Perception Revised', pp. 14–15. Constancy in speech perception is a complex phenomenon; we should be wary of assuming that speech perception exhibits constancy in just the sense in which colour perception does. For one thing, changes in how phones are produced can increase the amount of cognitive effort required to comprehend speech and can hinder or facilitate learning in different situations: Nygaard and Pisoni, 'Speech Perception', pp. 69ff. Constancy in speech perception may sometimes rely on limited cognitive resources such as attention and working memory.

[25] See Harnad (ed.), *Categorical Perception*. There is controversy over exactly how categorical perception should be defined (see B. Schouten *et al.*, 'The End of Categorical Perception as We Know It', *Speech Communication*, 41 (2003), pp. 71–80), and even on whether speech perception is categorical (e.g., D.W. Massaro and M.M. Cohen, 'Categorical or Continuous Speech Perception: a New Test', *Speech Communication*, 2 (1983), pp. 15–35). It would be a major undertaking to give a rigorous, plausible and useful characterization of the phenomenon; the present rough characterization (like that of Schouten *et al.*) is inadequate because it refers to discriminability of stimuli outright rather than to discriminability by means of a particular perceptual modality.

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least as well as to intended phonic gestures. If such a correspondence were found, it might be possible to give an explanation better than Liberman and Mattingly's for the existence of categories corresponding to intended phonic gestures.[26] Reasons for doubting any such explanation exists include the constancy effects already mentioned, and also co-articulation, the fact that phonic gestures overlap (this is what makes talking fast).

In outline, Liberman and Mattingly's argument for the claim that the objects of speech perception are intended phonic gestures has this form:

1. There are category boundaries
2. These boundaries correspond to intended phonic gestures
3. Facts (1) and (2) stand in need of explanation
4. The best explanation of (1) and (2) involves the claim that the objects of speech perception are intended phonic gestures.

This illustrates how one might establish claims about the objects of perception without relying too directly on what people say they perceive, and without presupposing very general theoretical claims about what it means for something to be perceived.

I am not suggesting that the case of speech perception can be taken as a model for understanding perception generally. In my view, speech perception, like other forms of categorical perception, is importantly different from ordinary, non-categorical perception. By contrast, some philosophers have developed general accounts of perception which appear to be supported primarily by cases of categorical perception such as speech and colour. For instance, both Fred Dretske and Jerry Fodor have taken speech perception as a paradigm case for very general theories about what it is for something to be an object of perception.[27] Mohan Matthen has recently gone even further by supposing that all perception is a matter of assigning stimuli to categories.[28] For reasons tentatively outlined later, these authors may be mistaken to take categorical forms of perception as paradigm cases of perception generally. The suggestion I develop next is only that the categorical perception of speech is a reasonable model for causal perception of visual stimuli.

[26] See Nygaard and Pisoni, 'Speech Perception', pp. 83–4, for a compact summary.

[27] F. Dretske, *Knowledge and the Flow of Information* (Oxford: Blackwell, 1981), pp. 166–7; J. Fodor and Z. Pylyshyn, 'How Direct Is Visual Perception? Some Reflections on Gibson's "Ecological Approach"', *Cognition*, 9 (1981), pp. 139–96, at p. 176.

[28] M. Matthen, *Seeing, Doing and Knowing* (Oxford: Clarendon Press, 2005).

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IV

Is speech perception a good model for causal perception? I suggest that it is, and that an argument like (1)–(4) above can be used to show that we perceive causal interactions.

The first step is to show that causal perception exhibits the key features of categorical perception. As already mentioned, whether or not the experience characteristic of launching occurs depends on interactions among various factors including the relative speeds of the two objects, the delay between the first and second objects' movements, and the trajectories of the two objects. Importantly, the difference between an experience characteristic of launching and an experience of two distinct motions often depends on small changes to parameters, changes smaller than can normally be discerned in perception. Figure 6 shows the effect of the interval between movements on whether subjects report experiencing direct launching or two distinct movements.[29] Subjects would not normally be able to distinguish stimuli between trials when the stimuli differ only in that the gap between movements is 28 ms longer on one of them; but they can easily discriminate stimuli in the special case where 28 ms make the difference between the experience characteristic of launching and the experience characteristic of two movements. As in the case of speech, then, there are points around which exceptionally fine-grained discriminations are possible. This is the key indicator that causal perception is categorical.

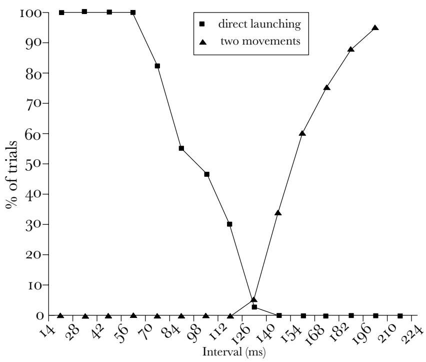

Figure 6

As well as such things as relative trajectories and speeds, the experience of launching also depends on high-level features of stimuli. For example, Michotte found (exp. 21, pp. 74ff.) that if the second ball in a launching sequence is initially seen moving backwards and forwards on its own account, then the experience distinctive of launching does not occur. So the experience characteristic of launching depends not only on simple spatial and temporal parameters but also on more complicated structural features

[29] Redrawn from Michotte, *The Perception of Causality*, p. 93, table IV.

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of the stimuli.[30] Further, Scholl and colleagues have shown that in some situations the experience characteristic of launching depends how events unfold shortly after the moment of collision.[31] This suggests that the relation between experiences and visual stimuli is complex in the same way as the relation between speech experiences and speech stimuli: in both cases there is no simple mapping from features of the stimuli to features of the experience unless one characterizes the stimuli in terms of gestures or causes.

In short, the experience characteristic of launching involves both the heightened discriminability and the sensitivity to complex features of stimuli which are characteristic of categorical perception. Research on other causal stimuli such as pulling and bursting is less well developed, but also provides some evidence that causal perception may be a form of categorical perception.[32]

To say that perceptual experiences characteristic of launching and other causal stimuli are categorical is not yet to say where the category boundaries fall. The experimental findings advanced so far rely on what subjects say about their experiences only as a way of distinguishing whether on two occasions they have the same or different types of experience. At this point, what the experiences are experiences of is an open question.

One can take a first step to answering this question by finding things the categories correspond to, that is, by enquiring which features of stimuli reliably predict whether they will be assigned to a given perceptual category. Of course this is only a first step. Facts about what perceptual categories correspond to do not straightforwardly determine what those categories are categories of.

Michotte argued (p. 223) that the conditions under which the experience characteristic of launching occurs correspond to physical laws governing collisions. In fact, he thought the correspondence so close that at one point (p. 228) he even suggested that it would be possible to learn about the physics of collisions by studying their perception:

[30] Saxe and Carey treat such findings as showing that subjects make judgements about whether an object is animate or inanimate: R. Saxe and S. Carey, 'The Perception of Causality in Infancy', *Acta Psychologica*, 123 (2006), pp. 144–65. Michotte, by contrast, argues (p. 68) that the effect is due to disrupting polarity, that is, disrupting the appearance of the first object approaching the second or the second object moving away from the first. Here I am supposing that Michotte is right to understand the effect as narrowly perceptual rather than as depending on judgements.

[31] See the contrast between 'dumbbell' and 'removal' conditions in Hoon Choi and B.J. Scholl, 'Effects of Grouping and Attention on the Perception of Causality', *Perception and Psychophysics*, 66 (2004), pp. 926–42, at p. 930.

[32] White and Milne (p. 512) cautiously advance the possibility that causal perception is categorical, noting that the experiments they conducted do not establish this.

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> ... anyone not very familiar with the procedure involved in framing the physical concepts of inertia, energy, conservation of energy, etc., might think that these concepts are simply derived from the data of immediate experience.

Opposing Michotte, Peter White and Elizabeth Milne suggest that the conditions under which the perceptions occur do not correspond to conditions required for actual pulling or bursting. Rather, White and Milne suggest (p. 515) that they may correspond to laws implicit in our naïve conceptions of causal interactions.

Evaluating these hypotheses requires painstaking empirical research on category boundaries (just as in the case of speech). To illustrate, Michotte found that the experience of launching occurs when there is a pause between two objects' movements of around 80 ms or less. How is this consistent with the laws of mechanics? Surely no pause can be tolerated? Ingeniously, Michotte (pp. 91–8, 124) compares launching with the movement of a single object. The single object moves half-way across a screen, then pauses before continuing to move. Michotte found that the longest pause between the two movements consistent with subjects' experiencing them as a single movement is around 80 ms, exactly the longest pause consistent with experiences characteristic of launching. Accordingly, the experience characteristic of launching appears to require that the two movements must be experienced as uninterrupted; this is why they can be separated by a pause of up to but no longer than 80 ms.

This is the kind of detailed investigation needed to support (or refute) claims that conditions for experiences characteristic of launching correspond to conditions for a certain kind of collision. Such research may eventually lead to fine-grained distinctions between competing hypotheses about which causal interactions are perceived and how they are perceived. As things stand, it is already reasonable to conclude that there is some correspondence between the conditions under which experiences characteristic of various types of causal interaction occur and conditions under which the causal interactions themselves occur.

This needs qualification, because few of the stimuli used in experiments on causal perception involve real causal interactions. Strictly speaking, then, it is false that conditions for experiences characteristic of launching correspond with conditions under which collisions occur. I need to hedge this claim by restricting it to the perceiver's natural environment, and excluding factors to which the relevant perceptual systems are insensitive.[33] In case this hedge seems like a fudge, exactly the same considerations apply in the case of speech. Earlier I mentioned that category boundaries in speech

[33] Michotte's 'paradoxical cases' may require further hedging (experiments 17, 70).

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perception correspond to differences in intended phonic gesture. But as speech perception is reliably fooled into treating all kinds of synthetic sounds as speech, this is not strictly true. As in the causal case, it is at best true taking into account the limits of perceptual systems, their natural environments and the types of information they are concerned with processing.

If a simple co-variational or causal theory of perceptual content were correct, the need to hedge claims about correspondence would show that we do not perceive causal interactions (and that we do not perceive intended phonic gestures). I am assuming that such theories are false, and I am pointing to correspondence not as a necessary or sufficient condition for perception but only as something to be explained. Given this, hedging would only be objectionable if done to such an extent that the correspondence in question were no longer a fact that needed explaining. After all, even imperfect correspondences need explanation (for example, correspondence between weather forecasts and actual weather conditions).

So far, then, I have covered the first three steps of an argument about causal perception parallel to the one about speech:

1′. There are category boundaries
2′. These boundaries correspond to certain types of causal interaction
3′. Facts (1′) and (2′) need explanation.

This brings me to the fourth and final step of the argument:

4′. The best explanation of (1′) and (2′) involves the claim that the objects of perception are causal interactions.

In defence of (4′), I conjecture that the perceptual experiences characteristic of causal interactions are best explained as by-products of perceptual mechanisms whose function is to track three-dimensional objects and their movements. Causal perception is part of object perception.

At least three pieces of evidence suggest that perception of objects is closely linked to causal perception. First, Michotte argued for such a link on the basis of his finding that launching occurs when there is a conflict between cues to object identity: good continuity of movement suggests a single object, whereas the existence of two distinct surfaces indicates two objects.[34] It is plausible that other types of causal interaction also involve conflicts between cues to object identity. Secondly, as I pointed out above, Scholl and Nakayama have shown that when a sequence involving complete overlap between two objects is perceived as a launching event, the subject perceives an illusory crescent as if the overlap were only partial. They

[34] 'Visual acuity favours, at one and the same time, the segregation of the objects and the unification of movements with the objects performing them' (Michotte, p. 51).

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conclude (p. 467) that 'the perception of causality can also affect other types of visual processing – in this case the perception of spatial relations among moving objects'. Thirdly, Krushke and Fragassi have shown that the object-specific preview effect vanishes in launching, but not in various spatio-temporally similar sequences.[35] Since the object-specific preview effect is diagnostic of feature binding, this is evidence that in launching sequences, features of the second object (such as motion) remain bound to the first object for a short time after the second object starts to move. (I do not have space to explain the object-specific preview effect here, but it is very clearly explained by Krushke and Fragassi.) These three pieces of evidence indicate that causal perception interacts with object perception. It is not that we first see sequences of events involving objects and then assign them causal significance. Rather, the perception of something as a causal interaction is bound up with what we perceive as objects in the first place.

This suggests a simple explanation for the correspondence between the perceptions characteristic of various kinds of causal interaction and conditions under which those causal interactions occur. The correspondence exists because given contingent facts about how it actually works, the perceptual system responsible for identifying objects must also concern itself with certain kinds of causal interaction in order to reconcile conflicting cues to object identity. In slightly more detail: one function of our perceptual systems is to identify and track objects; this is done by means of various cues; sometimes the visual system is faced with conflicting cues to object identity which need to be resolved in order to arrive at a satisfactory interpretation; when certain types of causal interaction occur there is a conflict among cues to object identity; these conflicts must be treated differently from other conflicts because they do not indicate failures of object identification, and so do not require resolution or further perceptual processing. So object perception depends on sensitivity to certain types of causal perception, and this is why perception of those types of causal interaction is categorical.

If this explanation is correct, one can conclude that causal interactions are objects of perception in the same sense as intended phonic gestures are. But this explanation is not as satisfying as the one offered in the case of speech: that explained *how* intended phonic gestures could be perceived, whereas this only says that causal interactions *need to be* perceived, given how object perception works.

Where can a deeper explanation be found? If, as I have been suggesting, object perception and causal perception are one and the same process, a

[35] J.K. Krushke and M.M. Fragassi, 'The Perception of Causality: Feature Binding in Interacting Objects', in *Proceedings of the Eighteenth Annual Conference of the Cognitive Science Society* (Hillsdale: Erlbaum, 1996), pp. 441–6.

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theory about how object perception works might also explain how causal interactions are perceived. Elizabeth Spelke's theory is that object perception is best understood as an inference where the premises are descriptions of stimuli in terms of surfaces and their layout; conclusions about objects are inferred from these premises by means of a set of principles describing how objects behave. Here, for example, are two of Spelke's four principles of object perception:

Boundedness. Two surface points lie on distinct objects only if no path of connected surface points links them ... the boundedness principle implies that ... two distinct objects cannot interpenetrate

No action at a distance. Separated objects are interpreted as moving independently of one another if such an interpretation exists.[36]

These principles bear not only on object identity but also on causal interactions between objects. In the case of launching, the no-action-at-a-distance principle is clearly relevant; for example, it would explain why there is no launching effect without spatial contact. The boundedness principle is also relevant, as it explains why the experience characteristic of launching is typically absent when there is complete overlap between two objects. In general, Spelke holds (p. 51) that 'object perception reflects basic constraints on the motions of physical bodies'.[37] Developing this idea, Renee Baillargeon, Su-hua Wang and their colleagues have argued that a single set of principles is responsible for object segmentation and for the perceptual identification of certain events as physically possible or impossible.[38] If this is right, the principles which explain object perception also explain how causal perception works.

This completes my argument for the claim that we can perceive certain types of causal interaction. This argument was outlined in (1′)–(4′) above. Clearly the argument is far from decisive, since it depends on a complex mix of theoretical and empirical issues. Even so, the argument demonstrates how to respond to Hume's claim that we do not perceive causal interactions. Given the evidence we have so far, it is more plausible that we perceive causal interactions than that we do not. Equally importantly, the comparison with

[36] E. Spelke, 'Principles of Object Perception', *Cognitive Science*, 14 (1990), pp. 49–50.

[37] See also S. Carey and E. Spelke, 'Domain-Specific Knowledge and Conceptual Change', in L. Hirschfeld and S. Gelman (eds), *Mapping the Mind: Domain Specificity in Cognition and Culture* (Cambridge UP, 1994), pp. 169–200, at p. 175: 'A single system of knowledge ... appears to underlie object perception and physical reasoning'.

[38] R. Baillargeon, 'Infants' Physical Knowledge: of Acquired Expectations and Core Principles', in E. Dupoux (ed.), *Language, Brain, and Cognitive Development: Essays in Honor of Jacques Mehler* (MIT Press, 2001), pp. 341–62; Su-hua Wang *et al.*, 'Detecting Continuity Violations in Infancy: a New Account and New Evidence from Covering and Tube Events', *Cognition*, 95 (2005), pp. 129–73.

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speech perception provides some insight into what it means to perceive causal interactions: they are perceived categorically, and they are objects of perception in the same sense as intended phonic gestures are.

Strictly speaking, this argument establishes only that we perceive causal interactions when presented with visual stimuli, not that we see them. Causal perception may be amodal, as speech perception is sometimes held to be.[39] Even so, I shall continue to write about seeing causes and hearing gestures for simplicity. Nothing said here contradicts David Marr's view (p. 36) that 'the quintessential fact of human vision – [is] that it tells about shape and space and spatial arrangement'.

V

So far I have argued that we can see some causal interactions, and that seeing causal interactions is part of seeing objects (seeing objects move is interdependent with, not prior to, seeing causal interactions). The argument followed a well understood model from the case of speech perception. First I pointed to a feature of causal perception in need of explanation, the existence of categories and their correspondence with situations in which causal interactions occur. Then I argued that the best explanation for this feature involves the hypothesis that we perceive causal interactions: the feature occurs because given how they actually work, perceptual systems responsible for identifying and tracking objects must also discriminate certain types of causal interaction.

As I mentioned at the start, many of those who agree that we can perceive causings also claim that we perceive causings in the same sense as we perceive shape, whatever sense that is. Full evaluation of this further claim would involve considering issues about perception beyond the scope of this

[39] The evidence on this point is controversial. Sekuler and colleagues show that when subjects observe an ambiguous visual display consistent with either a collision or a passing event, the timing of a tone can control whether subjects report seeing a collision or passing, and argue that this is a multisensory phenomenon: R. Sekuler *et al.*, 'Sound Alters Visual Motion Perception', *Nature*, 385 (1997), p. 308. Watanabe and Shimojo extend this finding by showing that not any event (or non-event) which draws attention at the moment of a collision will disambiguate the display; they argue that the tone's effect on the perception of a collision is a 'genuine audiovisual effect, not an audiovisual effect that results from auditory effects': K. Watanabe and S. Shimojo, 'When Sound Affects Vision: Effects of Auditory Grouping on Visual Motion Perception', *Psychological Science*, 12 (2001), pp. 109–16. Guski and Troje, on the other hand, show that features which carry no information about causation, such as a blink, can also influence whether subjects report seeing a collision or a passing. These authors conclude that auditory influences on the perceptual of collisions are 'no true cross-modal phenomenon': R. Guski and N.F. Troje, 'Audiovisual Phenomenal Causality', *Perception and Psychophysics*, 65 (2003), pp. 789–800, at p. 799.

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paper. On the other hand, some evaluation of this claim is necessary for understanding what it means to perceive causings. In a preliminary and tentative way, I suggest that the further claim is false, and I conclude with some considerations distinguishing causal and speech perception from shape perception.

Here are two claims about shape perception. First, in a wide range of naturally occurring circumstances, perceiving an object's shape is a way of being aware of its shape, that is, of being in a position to think about and act on its shape.[40] Secondly, to describe what someone perceives as including a particular shape is to describe a characteristic of their experience which is, often enough, introspectable.[41] I assume that these claims are true and constitutive of shape perception, and I argue that parallel claims about causal or speech perception are false. If these claims are indeed constitutive of shape perception but false of causal perception, then we do not perceive causal interactions in the same sense as we perceive shape.

When it comes to the relation between facts about the objects of speech perception and its introspectable character, not even the best trained, most conceptually sophisticated experts can use introspection to resolve conflicts about whether the objects of speech perception are sounds or gestures. The issues that determine the conflict concern the nature of the stimuli (what invariants are there?) and processes (does passive speech perception involve the motor cortex?). I suspect introspection tells us nothing about what we perceive when we perceive speech; certainly it has no bearing on conflicts about the objects of speech perception. So speech and shape perception differ with respect to introspection.

The parallel point about causal perception is less clear, because some philosophers and psychologists seem to have held that introspection can reveal what we perceive when we perceive causal interactions. While in principle there might also be perceptual experiences about whose objects facts are knowable through introspection, the type of perceptions which Michotte studied and which have been the focus of this paper are not like this. As in the case of speech perception, discerning the objects of causal perception depends on careful investigation of the conditions under which they occur and the reasons why they occur under these conditions. Introspection is inessential.

[40] M.G.F. Martin, 'Perception, Concepts and Memory', *Philosophical Review*, 101 (1992), pp. 745–63, at p. 761: 'Perception and experience ... are a matter of the world making itself apparent to us'.

[41] 'Describing [Mary's experience] as being as of a dodecahedron ... is ... normally intended to describe its introspectable character, that it is of how the physical world appears to be' (Martin, p. 762).

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What about the relation between perception and awareness? In the case of speech, researchers distinguish categorical perception of phonemes from 'phonological awareness', the ability to manipulate phonemes intentionally and reason about them. Phonological awareness is measured by a range of tasks that require sorting words according to their initial phoneme, segmenting or completing words, and blending or eliding phonemes.[42] Although these tasks might seem to test obscure abilities, phonological awareness has practical value in acquiring literacy.

Several factors jointly make it necessary to distinguish perception from awareness of phonemes. Most importantly, infants from four months or earlier enjoy categorical perception of phonemes, but it takes them until around four years before they can think or reason about phonemes.[43] Infants and younger children who hear 'sat' or 'bat' can distinguish the words by virtue of perceiving distinct intended phonic gestures but cannot act on or think about this difference.[44] Perception and awareness also differ with respect to the way they develop and the factors that influence their development. Unlike phoneme perception, children's acquisition of phonological awareness develops gradually over several years, varies systematically depending on their oral language (e.g., Turkish *vs* French), and is facilitated by learning a writing system where some types of writing system help more than others (e.g., syllabaries *vs* alphabets). Furthermore, children find certain types of phoneme harder to distinguish in thought than others (e.g., those

[42] There is a short review in J.L. Anthony and C.J. Lonigan, 'The Nature of Phonological Awareness: Converging Evidence from Four Studies of Preschool and Early Grade School Children', *Journal of Educational Psychology*, 96 (2004), pp. 53–5. While children typically pass different tests for phoneme awareness at slightly different times, they are collectively taken to measure phonological awareness because longitudinal surveys have shown that success or failure on these tasks is best explained by a single factor: see J.L. Anthony and D.J. Francis, 'Development of Phonological Awareness', *Current Directions in Psychological Science*, 14 (2005), pp. 255–9, at p. 256.

[43] Eimas and colleagues trained infants to suck in order to hear a sound and were then able to identify infants' interest in different sounds by measuring how vigorously they suck in order to hear the next sound. Since novel sounds are more interesting than familiar sounds, this experiment reveals which sounds infants regard as different and which they treat as the same. The headline finding is that four-month-olds have categorical perception of some phonemes: P.D. Eimas *et al.*, 'Speech Perception in Infants', *Science*, 171 (1971), pp. 303–6. Further research established that infants perceive many phonemes much as adults do. See P. Jusczyk, 'Language Acquisition: Speech Sounds and the Beginning of Phonology', in J. Miller *et al.* (eds), *Speech, Language and Communication* (San Diego: Academic Press, 1995), pp. 263–301, at p. 267: 'from a very early age, infants discriminate many, if not all, of the contrasts that are likely to occur among words in the native language. Moreover, at least on a general level, there are some striking similarities between the way that infants and adults respond to the same kinds of speech contrasts.'

[44] This was first noted by I.Y. Liberman *et al.*, 'Explicit Syllable and Phoneme Segmentation in the Young Child', *Journal of Experimental Child Psychology*, 18 (1974), pp. 201–12, at p. 203: 'it does not follow from the fact that a child can easily distinguish *bud* from *bat* that he can therefore respond analytically to the phonemic structure that underlies the distinction'.

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that differ only with respect to voicing are harder to distinguish than those differing only with respect to articulation), whereas they have no corresponding difficulties perceiving distinctions between phonemes. In principle there might be several ways of explaining these developmental findings. But there is a natural explanation which is generally taken for granted by researchers in this area: in becoming aware of phonemes we have to rediscover them, lacking access to them as objects of perception; and we become aware of phonemes as things which sometimes differ with respect to their identities and properties from the intended phonic gestures we perceive.

If this is correct, phoneme and shape perception differ in that perceiving phonemes is not a way of being aware of them. What about causal perception? As in the case of speech, there is a long gap between perceiving and the onset of reasoning abilities. Infants enjoy categorical perception of causal interactions from no later than nine months,[45] whereas children are two and a half or three years old before they can engage in even the simplest forms of causal reasoning.[46] As in the case of speech, children's ability to make judgements about visually presented causal interactions appears to improve gradually over several years up to around five years old,[47] and they have difficulty integrating perceptual information with their reasoning.[48] In the causal case, some investigators regard perception and awareness as closely connected despite the developmental gap, arguing that it can be explained by extraneous factors such as methodological deficiencies or task demands.[49] Given the uncertainty in this area, I suggest that the points of comparison

[45] Some of the first experiments with infants appear in A. Leslie and S. Keeble, 'Do Six-Month-Old Infants Perceive Causality?', *Cognition*, 25 (1987), pp. 265–88; L.M. Oakes and L.B. Cohen, 'Infant Perception of a Causal Event', *Cognitive Development*, 5 (1990), pp. 193–207. For more recent surveys see L.B. Cohen *et al.*, 'The Development of Infant Causal Perception', in A. Slater (ed.), *Perceptual Development: Visual, Auditory, and Speech Perception in Infancy* (Hove: Psychology Press, 1998), pp. 167–210; Saxe and Carey, 'The Perception of Causality in Infancy'.

[46] See, e.g., N.E. Berthier *et al.*, 'Where's the Ball? Two- and Three-Year-Olds Reason about Unseen Events', *Developmental Psychology*, 36 (2000), pp. 394–401. Bruce Hood and colleagues directly contrasted causal tasks requiring perceptual responses with parallel tasks requiring simple reasoning: B. Hood *et al.*, 'Looking and Search Measures of Object Knowledge in Preschool Children', *Developmental Psychology*, 39 (2003), pp. 61–70.

[47] A. Schlottmann *et al.*, 'Perceptual Causality in Children', *Child Development*, 73 (2002), pp. 1656–77, at p. 1671.

[48] A. Schlottmann, 'Perception Versus Knowledge of Cause and Effect in Children: When Seeing Is Believing', *Current Directions in Psychological Science*, 10 (2001), pp. 111–15, and 'Seeing it Happen and Knowing How it Works: How Children Understand the Relation between Perceptual Causality and Underlying Mechanism', *Developmental Psychology*, 35 (1999), pp. 303–17.

[49] M. Haith, 'Who Put the Cog in Infant Cognition? Is Rich Interpretation Too Costly?', *Infant Behaviour and Development*, 21 (1998), pp. 167–79; R. Keen, 'Representation of Objects and Events: Why Do Infants Look So Smart and Toddlers Look So Dumb?', *Current Directions in Psychological Science*, 12 (2003), pp. 79–83.

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between speech and causal perception make plausible as a contender the alternative hypothesis that the developmental gap between perception and awareness is explained by the fact that causal perception is not a way of being aware of causal interactions.

These considerations indicate that causal perception differs from shape perception with respect to introspection and awareness. The role of categorical perception of causings is only to distinguish events, not to make manifest to us how the events are distinguished or the natures of the events so distinguished. From the point of view of the perceiver, categorical perception is simply a signal of sameness or difference with distinctive phenomenological effects. In this respect, causal perception is comparable to sensation: perceiving causal interactions is no more or less a way of being aware of them than experiencing a distinctive stinging sensation is a way of being aware of a nettle. If this is right, causal perception may also resemble sensation with respect to epistemology: seeing causings is not a way of gaining knowledge about them, and can be a source of knowledge only because we learn to associate characteristic phenomenology with the kind of events which give rise to it. Certainly we should reject the claim that causal perception is a source of causal concepts.[50]

In the philosophy of perception there is a divide between those who take the claims about introspection and awareness as defining the phenomenon and those who offer theories which are at odds with, or at least hard to reconcile with, these claims. Fred Dretske's views illustrate the second school of thought:

> Seeing objects is a way of getting information about them.... What makes it X (rather than Y) that we see is that the information these internal events carry is information about X (rather than Y).[51]

On the face of it, an internal event's carrying information about something does not necessarily involve either our being aware of that thing or the event's having a particular introspective character. Something like Dretske's way of thinking about perception might be true to causal and speech perception, and perhaps categorical perception more generally. But since these forms of perception may differ from shape perception with respect to awareness and introspection, it does not follow that the same approach is true to shape perception. Equally, although the first view cannot be true to causal or speech perception, because on that view it makes no sense to distinguish sharply perception of an object from awareness of it (on this

[50] This claim is made by Michotte, pp. 270–1, Ducasse, 'Causation: Perceivable? Or Only Inferred?', p. 177, and Armstrong, 'Going through the Open Door Again'.

[51] Dretske, 'Simple Seeing', in his *Perception, Knowledge and Belief* (Cambridge UP, 2000), p. 112.

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view, perceiving something sometimes just is being aware of it), it does not follow that the view is false when restricted to non-categorical forms of perception.

In conclusion, we do perceive causal interactions, and doing so is part of perceiving objects. In this respect Hume was wrong, but he may have been more right than wrong. For perceiving causal interactions may differ from perceiving shape with respect to how such perceptions relate to awareness and introspection. If these relations are defining characteristics of the kind of perception involved in seeing shapes, Michotte, Strawson and Scholl *et al.* are wrong to regard causal and shape perception as of a kind. We see causal interactions in the same sense as that in which we hear intended phonic gestures.[52]

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